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THE USE OF SURVEY WEIGHTS IN MACHINE LEARNING

With an application to prevalence estimation

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Table of Contents

Contents

Abstract	3
1. Introduction	4
2. General settings and notations	5
2.1 Sampling designs	5
2.2 Population prevalence	6
3. Prevalence estimation methods	7
3.1 Direct prevalence estimation	8
3.2 Logistic model	9
3.3 Logistic GREG estimators.....	10
3.4 Neural networks estimation.....	12
3.4.1 Fitting a neural network	14
3.4.2 Point and interval estimates	14
3.5 Model-assisted ANN with difference estimator	15
4. Simulation settings	17
5. Simulation results and comparisons	18
6. Concluding Remarks.....	23
7. References	24
8. Appendix.....	26
8.1 The variables in the simulated population:	26
8.2 Histogram of age by sex	26
8.3 Tables.....	27
8.2 R programs.....	

Abstract

Often, statisticians work on surveys that sample different population segments with different probabilities. Surveys are designed this way to obtain more precise information on the smaller subgroups in the population. Complex surveys typically involve clustering, stratification or both. However, complex surveys require more careful planning, design, and analysis than simple random samples. Sampling weights (the inverse probabilities of selection for each observation) allow us to reconfigure the sample as if it were a simple random draw of the total population and yield accurate population estimates for the main parameters of interest. The use of sampling weight and survey design is often focused on the traditional statistics models.

Due to the advanced computational capacity, machine learning has become popular and is often applied in complex surveys. However, there is a lack of discussion on whether the survey design should be incorporated into the learning algorithms. As such, this report compares traditional statistical models and machine learning algorithms in model-based and model-assisted approaches to using data from a complex survey to estimate population prevalence. We investigate the performance of the estimators through simulation studies.

Keywords: survey weights, stratified sampling, neural network, logistic regression, prevalence, model-assisted estimation methods, simulations.

1. Introduction

The use of survey weighting is one of the critical issues in surveys. Surveys are used to gather data on a small subset of a population to estimate finite population parameters, describing some aspect of the finite population under study. Most often, the data collected are stored in a rectangular data file, each row corresponding to a sample unit (e.g., a business, a household, an individual, etc.) and each column corresponding to a characteristic of interest (age, gender, income, etc.). Made available on the data file is a column of final weights. This set of weights constitutes a weighting system. The idea is to construct a single weighting system applicable to all the characteristics of interest.

The cornerstone of the foundation for design-based inference in survey sampling was laid out by Neyman (1934). Under the design-based framework, the survey population is a fixed finite set of N units. Values of the response variable y and covariates x associated with a particular unit are also viewed as fixed. Another approach Cochran (1939) proposed to analyze survey data is using statistical models. It is referred to as a model-based approach, assuming the finite population values, x and y are random samples from a superpopulation model, ξ . These two different approaches have been a subject of debate among survey statisticians (Wu & Thompson, 2020). The model-assisted approach is, then, suggested to incorporate the strength of plausible models into the design-based inference for survey samples. However, there is limited literature on how design-based inference should be incorporated when using machine learning algorithms on survey data.

In this project, we compare these approaches to prevalence estimation. For the design-based approach, we directly estimate the prevalence using the Horvitz - Thompson estimator (Horvitz and Thompson(1952)). When we use the model-based approach to estimate, we select logistic regression and artificial neural network (ANN) as methods representing statistical and machine learning, respectively. We select generalized logistic regression and ANN difference estimator as the model-assisted counterparts to estimate the population prevalence.

This report is organized as follows: in Section 2, we specify the details of the population and the sample designs. We then describe the point and interval estimates of the mentioned methods for population prevalence. Then, we present the simulation settings in Section 4 and discuss the simulation results in Section 5. Lastly, we conclude the study and provide some future research insight in Section 6

2. General settings and notations

We consider an inferential framework common among literature (Godambe and Thompson 1986, Breidt and Opsomer 2017, So, Thompson, and Wu 2020). The framework includes two sources of variations: the probability sampling design for a finite population and the model assumed for the responses for a stochastic superpopulation, ξ . For the inferential propose, we view the finite populations $U_{(\nu)}$, of the population sizes $N_{(\nu)}, \nu \rightarrow \infty$ as a sequence of realization from the superpopulation. The corresponding samples, $S_{(\nu)}$, have the sample sizes $n_{(\nu)}$. Both $N_{(\nu)}$ and $n_{(\nu)}$ go to infinity as $\nu \rightarrow \infty$, and $\frac{n_{(\nu)}}{N_{(\nu)}} \rightarrow \pi^* \in (0,1)$ (Breidt & Opsomer, 2017). For brevity reasons, we suppress the subscript ν and obtain the asymptotic results based on $N \rightarrow \infty$ or $n \rightarrow \infty$.

2.1 Sampling designs

Let U be a finite population realized from the superpopulation and consist of N units. In survey sampling, the interest usually lies in estimating finite population parameters such as the population total (or the population mean) of a characteristic of interest. Let $x_1, \dots, x_q$, denote q characteristics of the sampling units (e.g., income, age, gender, job status, etc.) and let $\mathbf{x}_k = (x_{0k} \equiv 1, x_{1k}, \dots, x_{qk})$ be the $(q + 1)$ -variate vector of variables corresponding to the k th sampling unit, where $k = 1, \dots, N$. We combine the $\mathbf{x}_1, \dots, \mathbf{x}_N$ to be $N \times (q + 1)$ matrix and denote it as X . . Sometimes, the variables in X are so common that most of the modern censuses collect such information on a regular basis. It implies that researchers know X for all the sampling units in the population, which is one of the assumptions in this research.

To estimate finite population parameters, it is a widespread practice to select a sample S from the finite population. The sampling design controls the random behavior of the sample and any estimators computed from it. Especially when the sampling design is complex, including one or several levels of random selection with unequal probabilities, it is essential to incorporate design information in constructing estimators for statistical validity and efficiency reasons. For this research, we focus on the stratified sampling framework like the one from Chapter 3 of (Lohr, 2009) widely adopted in health surveys.

In this project, we assume that there are H strata and there are $N_1, \dots, N_H$ population units in strata U_1 to U_H such that $N = \sum_{h=1}^H N_h$ and $U = \bigcup_{h=1}^H U_h$. We also define that S_h to be the simple random sample without replacement (SRSWOR) from the h th stratum with n_h sampled units. The sample can be written as $S = \bigcup_{h=1}^H S_h$ and the total sample size would be $n = n_1 + \dots + n_H$. Under the SRSWOR assumption, the sample weight of the k th unit would be $w_k = N_h/n_h$, if $k \in U_h$, which is the receptacle of the first-order inclusion probability $\pi_k = 1/w_k = \Pr(I_k = 1 | k \in U_h)$, and $I_k = 1$ if the k -th unit is selected in the sample S . Under this sampling design, it is straightforward to see the second-order inclusion probability as

$$\begin{aligned} \Pr(I_k = 1, I_l = 1 | k \in S_{h'}, l \in S_h) &= \pi_{kl} \\ &= \begin{cases} n_h(n_h - 1)/N_h(N_h - 1) & \text{if } h' = h, \text{ and } k \neq l \\ n_h n_{h'}/(N_h N_{h'}) & \text{if } h' \neq h \end{cases} \end{aligned}$$

Typically, design weights are used to compensate for over- or under-sampling of specific cases or disproportionate stratification. The design weights are used when we want the statistics to represent the population. (Chen et al., 2017; Haziza & Beaumont, 2017). Let x_k be a vector of auxiliary variables. In our sample, the x_k are known and sample vectors $\{x_k\}_{k \in S}$ are observed. Let us also assume a setting where some estimators require the stronger condition that the individual vectors $\{x_k\}_{k \in U}$ are known for the entire finite population.

2.2 Population prevalence

Let y denote a characteristic of interest, for example, the indicator of a particular disease or medical condition. The occurrence of a disease may often relate to numerous attributes such as age, gender, and sex. Those characteristics are often collected in a survey as X , as we defined in the previous section. Therefore, it is reasonable to assume that the dichotomous response Y to be random under the superpopulation ξ , that

$$\Pr(Y = 1 | x) = \mu(x) \tag{1}$$

where $\mu(\cdot)$ is the link function of the covariate x . In the finite population U , the realized response variables are $y_1, \dots, y_N$. The link function can be any function that gives a value between 0 and 1, and it is unknown in most cases.

The critical parameter, population total, is given as $t_y = \sum_{k \in U} y_k$, where y_k denotes the k th value of the response variable y , $k = 1, \dots, N$. Usually, we are

interested in the proportion of the population who have medical conditions. The prevalence can simply be calculated as the

$$p_y = \frac{t_y}{N} = \frac{1}{N} \sum_{k \in U} y_k$$

In stratum h , we can also define the stratum total as $t_{y,h} = \sum_{k \in U_h} y_k$ and the prevalence of the stratum h can, therefore, be defined as

$$p_{y,h} = \frac{t_{y,h}}{N_h} = \frac{1}{N_h} \sum_{k \in U_h} y_k,$$

for $h = 1, \dots, H$. Although estimating the stratum prevalence is common, our research focuses on the population prevalence, p_y , when comparing the performance of various estimation methods. Most researchers put more emphasis on the population prevalence rather than the prevalence in strata, for example (Eide et al., 2015; Liu et al., 2014; Sutherland & Shepherd, 2001)

3. Prevalence estimation methods

When estimating the prevalence, if we ignore the sampling design as well as the covariates X , we can define the unweighted prevalence estimator as simply as

$$\hat{p}_y^{(un)} = \frac{1}{N} \frac{N}{n} \sum_{k \in S} y_k = \frac{1}{n} \sum_{k \in U} y_k I_k,$$

However, the unweighted estimator is design-biased,

$$E\left(\hat{p}_y^{(un)}\right) = \frac{1}{n} \sum_{k \in U} \frac{y_k n_k}{N_h} \neq \frac{1}{n} \sum_{k \in U} y_k.$$

It motivates us to search for different methods to estimate population prevalence. Those methods should consider the survey designs or the other covariates and supplementary information, the survey provides.

As the link function $\mu(\cdot)$ in (1) cannot be identified, we consider statistical learning algorithms, like logistic regression or machine learning algorithms, like artificial neural network (ANN), to approximate the unknown link function.

This project emphasizes the naïve method, which directly estimates the population prevalence, logistic generalized regression (GREG) estimators, and neural network prediction method.

3.1 Direct prevalence estimation

One basic and intuitive way to estimate the population prevalence is to estimate the average number of people having the disease conditions in the population. When the total population N , is known, we estimate the prevalence by the Horvitz - Thompson estimator,

$$\hat{p}_y^{(D)} = \frac{1}{N} \sum_{k \in S} w_k y_k = \frac{1}{N} \sum_{k \in U} w_k y_k I_k ,$$

where y_k is the indicator of the k th unit defined as

$$y_k = \begin{cases} 1 & \text{if the } k\text{th unit has the disease} \\ 0 & \text{otherwise} \end{cases} .$$

We immediately know that it is unbiased, as

$$E \left(\hat{p}_y^{(D)} \right) = \sum_{k \in U} \frac{y_k}{N} , \quad E \left(\frac{I_k}{\pi_k} \right) = \sum_{k \in U} \frac{y_k}{N} .$$

The standard error of the direct prevalence estimator and $SE \left(\hat{p}_y^{(D)} \right)$ is

$$\left(SE \left(\hat{p}_y^{(D)} \right) \right)^2 = \frac{1}{N^2} \sum_{k, l \in U} Cov(I_k, I_l) \frac{y_k y_l}{\pi_k \pi_l} = \frac{1}{N^2} \sum_{k, l \in U} \Delta_{kl} \frac{y_k y_l}{\pi_k \pi_l} , \quad (2)$$

where $\Delta_{kl} = \pi_{kl} - \pi_k \pi_l$. There are many options to estimate the standard error, but in this study, we choose the old-fashioned design-based unbiased variance estimator,

$$\left(\widehat{SE} \left(\hat{p}_y^{(D)} \right) \right)^2 = \frac{1}{N^2} \sum_{k, l \in S} \frac{\Delta_{kl} y_k y_l}{\pi_{kl} \pi_k \pi_l} .$$

Under stratified sampling, if $k \in S_{h'}$ and $l \in S_h$, we have

$$\Delta_{kl} = \begin{cases} \frac{n_h(n_h - 1)}{N_h(N_h - 1)} - \left(\frac{n_h}{N_h} \right)^2 & \text{if } h' = h \\ 0 & \text{if } h' \neq h \end{cases}$$

Finally, we can derive the Wald confidence interval to be

$$\hat{p}_y^{(D)} \pm z_{1-\alpha/2} \widehat{SE}(\hat{p}_y^{(D)}),$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ quantile of a standard normal distribution.

We can easily implement this method with the `svymean` function of the `survey` package in R. We provide the detailed R codes in the Appendix.

It is a purely *design-based* approach which assumes that the randomness in the estimators is due only to the random selection of the sample, see (Breidt & Opsomer (2017), Wu & Thompson (2020)). Generally speaking, $\hat{p}_y^{(D)}$ tends to be unstable, especially when the π_k 's are highly dispersed such that some π_k 's are remarkably close to zero. In this case, $\hat{p}_y^{(D)}$ would be poorly related to the characteristic of interest and could not provide a reliable estimation. Besides, this method does not consider the relationship between the covariates and the disease indicators. There is a huge information loss and the prevalence estimates of the finite population would be highly inaccurate.

3.2 Logistic model

To incorporate the covariate information, we can use a parametric model link relationship between x and y . The most popular model for the binary response is the logistic regression model (Agresti (2019)). In the model, we assume the logit of the disease probability to be a linear function of the covariate $\mathbf{x}_k = (1, x_{k1}, \dots, x_{kq})^T$,

$$\text{logit}(\mu(\mathbf{x}_k)) = \log\left(\frac{\mu(\mathbf{x}_k)}{1 - \mu(\mathbf{x}_k)}\right) = \beta_0 + \beta_1 x_{1k} + \dots + \beta_q x_{qk} = \mathbf{x}_k^T \boldsymbol{\beta},$$

where $\mu(\mathbf{x})$ is defined in (1), and $\boldsymbol{\beta}$ is $(q + 1)$ -variate vector of parameters where the β_j is the log odds of $y = 1$ for every 1- unit increase in x_j , for $j = 1, \dots, q$. We can estimate $\boldsymbol{\beta}$ by maximizing the log-likelihood function,

$$\ell(\boldsymbol{\beta}) = \sum_{k \in S} [y_k (\mathbf{x}_k^T \boldsymbol{\beta}) - \ln(1 + \exp(\mathbf{x}_k^T \boldsymbol{\beta}))] \quad (3)$$

and obtain the maximum likelihood estimators $\widehat{\boldsymbol{\beta}}^{(logit)}$, respectively. The prevalence can be estimated as $\hat{p}_y^{(logit)} = \hat{p}_y(\widehat{\boldsymbol{\beta}}^{(logit)})$, where

$$\hat{p}_y(\boldsymbol{\beta}) = \frac{1}{N} \sum_{k \in U \setminus S} \frac{\exp(\mathbf{x}_k^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_k^T \boldsymbol{\beta})} + \frac{1}{N} \sum_{k \in S} y_k \quad (4)$$

The standard error of $\hat{p}_y^{(logit)}$ can be obtained by the delta's method,

$$\left(SE \left(\hat{p}_y^{(logit)} \right) \right)^2 \approx \hat{p}' \left(\hat{\beta}^{(logit)} \right) V_{\hat{\beta}}^{(logit)} \left(\hat{p}' \left(\hat{\beta}^{(logit)} \right) \right)^T,$$

where $\hat{p}'(\beta) = \left(\sum_{k \in U \setminus S} \frac{x_k^T}{N} \frac{\exp(x_k^T \beta)}{(1 + \exp(x_k^T \beta))^2} \right)$ and $V_{\hat{\beta}}^{(logit)}$ is the variance-covariance

matrix of $\hat{\beta}^{(logit)}$. It is a frequent practice to estimate $\hat{V}_{\hat{\beta}}^{(logit)}$ by the inverse of the observed information matrix,

$$\hat{V}_{\hat{\beta}}^{(logit)} = I_{obs}^{-1} \left(\hat{\beta}^{(logit)} \right) = \left[-\frac{\partial^2 \ell}{\partial \beta^2} \right]_{\beta = \hat{\beta}^{(logit)}}^{-1}$$

Therefore, we can obtain a $(1 - \alpha)100\%$ Wald confidence interval for $\hat{p}_y^{(logit)}$

$$\hat{p}_y^{(logit)} \pm z_{1-\alpha/2} \widehat{SE} \left(\hat{p}_y^{(logit)} \right),$$

where $\widehat{SE} \left(\hat{p}_y^{(logit)} \right)$ is the estimate of the standard error,

$$\widehat{SE} \left(\hat{p}_y^{(logit)} \right) = \sqrt{\hat{p}'_y \left(\hat{\beta}^{(logit)} \right) \hat{V}_{\hat{\beta}}^{(logit)} \left(\hat{p}'_y \left(\hat{\beta}^{(logit)} \right) \right)^T}.$$

Note that this method does not incorporate the survey design. It is a model-based design that ignores the information given by the survey design. It treats (y_k, x_k) as a realized “sample” from the superpopulation model ξ . The uncertainty of the estimation is solely from the randomness of the variable y in ξ , rather than the randomness from the probability sampling design (Wu & Thompson, 2020).

3.3 Logistic GREG estimators

To incorporate the survey design into the logistic regression model, we consider the model-assisted approach, which considers sampling weights in model parameter estimation. We have a similar model setup as the logistic regression in Section 3.2, but the likelihood function (3) becomes the pseudo log-likelihood function (Lohr, n.d.)

$$\ell^{(GREG)}(\beta) = \sum_{k \in S} w_k [y_k(x_k^T \beta) - \ln(1 + \exp(x_k^T \beta))]]$$

The $\hat{\beta}^{(GREG)}$ would be the pseudo maximum likelihood estimator, which solves the score functions

$$S(\boldsymbol{\beta}) = \sum_{k \in S} w_k \mathbf{x}_k \left[y_k - \frac{\exp(\mathbf{x}_k^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_k^T \boldsymbol{\beta})} \right] = \sum_{k \in S} w_k S_k(\boldsymbol{\beta}) = \mathbf{0}$$

Similar to Section 3.1, the prevalence can be estimated as $\hat{p}_y^{(GREG)} = \hat{p}_y(\hat{\boldsymbol{\beta}}^{(GREG)})$, where $\hat{p}_y(\boldsymbol{\beta})$ is defined in (3.1), and the standard error of $\hat{p}_y^{(GREG)}$ can be obtained by the delta's method,

$$\left(SE \left(\hat{p}_y^{(GREG)} \right) \right)^2 \approx \hat{p}'(\hat{\boldsymbol{\beta}}^{(GREG)}) V_{\hat{\boldsymbol{\beta}}}^{(ma)} \left(\hat{p}'(\hat{\boldsymbol{\beta}}^{(GREG)}) \right)^T,$$

However, unlike Section 3.1, we need some extra steps to estimate $V_{\hat{\boldsymbol{\beta}}}^{(GREG)}$. Consider,

$$Var_{\xi\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) = Var_{\xi} \left(E_{\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) \right) + E_{\xi} \left(Var_{\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) \right)$$

where $Var_{\xi} \left(E_{\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) \right)$ is known as the “model variance component,” and $E_{\xi} \left(Var_{\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) \right)$ is called the “design variance component” or the “sample variance component.” As $\hat{\boldsymbol{\beta}}$ is the solution of $S(\boldsymbol{\beta})$, $E_{\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) \approx 0$, and the model variance component becomes zero (If the sampling fraction is small, these quantities would be close to zero.) Therefore,

$$Var_{\xi\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) = E_{\xi} \left(Var_{\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) \right)$$

We can also consider the score function $S_j(\hat{\boldsymbol{\beta}})$ as the weighted sum of $S_{kj}(\hat{\boldsymbol{\beta}})$, and we can deduce its design variance, which is the variance due to the sample design, like (2), we have

$$\widehat{Var}_{\xi\pi} \left(S(\hat{\boldsymbol{\beta}}) \right) \approx \sum_{k,l \in S} \frac{\Delta_{kl}}{\pi_{kl}} \frac{S_k(\hat{\boldsymbol{\beta}})}{\pi_k} \frac{[S_l(\hat{\boldsymbol{\beta}})]^T}{\pi_l}.$$

Readers may refer to Särndal et al. (1992) for the theory of model-assisted estimation on the generalized model-based logistic regression model. Consider the Taylor series expansion at the population parameter $\boldsymbol{\beta}_N$, which is the solution of $\sum_{k \in U} S_k(\boldsymbol{\beta}) = \mathbf{0}$

$$S(\boldsymbol{\beta}_N) = S(\hat{\boldsymbol{\beta}}^{(GREG)}) + \left[\frac{\partial^2 \ell}{\partial \boldsymbol{\beta}^2} \right]_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}^{(ma)}} (\boldsymbol{\beta}_N - \hat{\boldsymbol{\beta}}^{(GREG)}) + o_p(n^{-1/2}).$$

Thus, it leads to the estimate of $Var(\hat{\beta}^{(GREG)})$

$$\widehat{Var}_{\xi\pi}(\hat{\beta}^{(GREG)}) \approx I_{obs}^{-1}(\hat{\beta}^{(GREG)}) Var_{\xi\pi}(S(\hat{\beta}^{(GREG)})) I_{obs}^{-1}(\hat{\beta}^{(GREG)}),$$

where $I_{obs}^{-1}(\hat{\beta}^{(GREG)}) = \left[-\frac{\partial^2 \ell}{\partial \beta^2} \right]_{\beta=\hat{\beta}^{(GREG)}}^{-1}$. Then, we can obtain,

$$\hat{V}_{\hat{\beta}}^{(GREG)} = \widehat{Var}_{\xi\pi}(\hat{\beta}^{(GREG)}) \approx I_{obs}^{-1}(\hat{\beta}^{(GREG)}) \widehat{Var}_{\xi\pi}(S(\hat{\beta}^{(ma)})) I_{obs}^{-1}(\hat{\beta}^{(GREG)}).$$

The $(1 - \alpha)100\%$ Wald confidence interval for $\hat{p}_y^{(GREG)}$

$$\hat{p}_y^{(GREG)} \pm z_{1-\alpha/2} \widehat{SE}(\hat{p}_y^{(GREG)}),$$

where $\widehat{SE}(\hat{p}_y^{(GREG)})$ is the estimate of the standard error,

$$\widehat{SE}(\hat{p}_y^{(GREG)}) = \sqrt{\hat{p}'(\hat{\beta}^{(GREG)}) \hat{V}_{\hat{\beta}}^{(GREG)} (\hat{p}'(\hat{\beta}^{(GREG)}))^T}.$$

3.4 Neural networks estimation

Artificial neural networks (ANN) are machine learning techniques that simulate learning mechanisms in living beings by computational units called neurons. (Aggarwal, 2018). Neural networks have two-stage regression or classification model. These networks can handle many quantitative responses seamlessly. In this project, the network has one output unit for binary classification, modeling the success probability (Hastie et al.). Denote z_m to be derived features created from linear combinations of the inputs (covariates) x_k and y_k is a function of linear combinations of the z_m ,

$$\begin{aligned} z_{mk} &= \sigma(\mathbf{a}_m^T \mathbf{x}_k), \quad m = 1 \dots M \\ T(\mathbf{x}_k) &= \mathbf{b}^T \mathbf{z}_k, \quad k = 1 \dots K \\ p(\mathbf{x}_k) &= g(T(\mathbf{x}_k)), \end{aligned} \tag{5}$$

where $\mathbf{a}_m = (a_{0m}, a_{1m}, \dots, a_{qm})^T$, $\mathbf{b} = (b_0, b_1, \dots, b_M)^T$ and $\mathbf{z}_k = (1, z_{1k}, z_{2k}, \dots, z_{Mk})^T$, the activation function $\sigma(\cdot)$ is a sigmoid function, $\sigma(v) = (1 + \exp(-v))^{-1}$.

We compare the similarity between the actual neuron and the artificial neuron in Figures 1 and 2. We also present the relationships of variables of our simple ANN in Figure 3.

FIGURE 1 BIOLOGICAL NEURON AND ARTIFICIAL NEURAL NETWORK¹

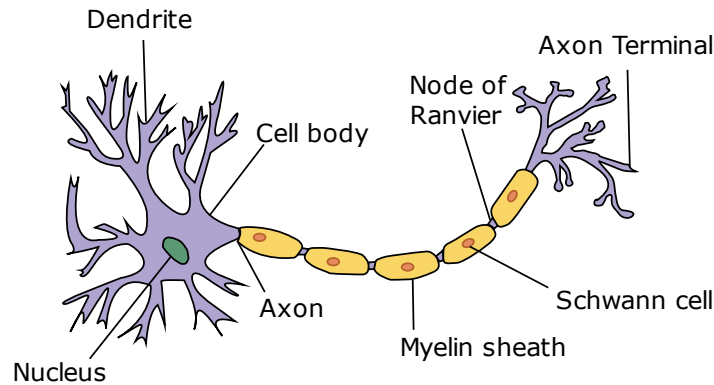


FIGURE 2: STRUCTURE OF A TYPICAL ARTIFICIAL NEURON

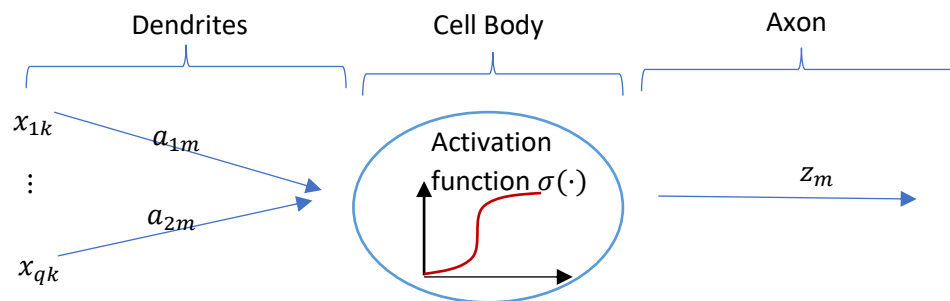
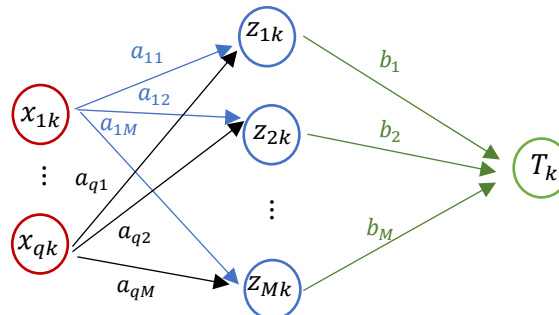


FIGURE3. A GRAPHICAL REPRESENTATION OF A SIMPLE ANN MODEL WE USED IN THIS PROJECT



The units in the middle of the network diagram which compute z_m (derived features) are called “hidden units” since the values z_m are not directly observed. The output function $g(T(x_k))$ allows a final transformation of the outputs vector $T(x_k)$. In binary classification, we take $g(\cdot)$ to be the softmax function,

$$g(t) = \frac{\exp(t)}{1 + \exp(t)},$$

which is the same logit function used in the logistic regression model. Figure 3 also has a bias unit feeding into every unit in the hidden and output layers, which captures the intercepts a_{0m} and b_0 in model.

¹ <https://upload.wikimedia.org/wikipedia/commons/b/b5/Neuron.svg>

3.4.1 Fitting a neural network

To obtain the unknown weights $\theta = \{\mathbf{a}_1, \dots, \mathbf{a}_m, \mathbf{b}\}$, where $\mathbf{a}_m = (a_{0m}, a_{1m}, \dots, a_{qm})^T$, $\mathbf{b} = (b_0, b_1, \dots, b_M)^T$, we minimize the following squared error or deviance function $R(\theta) = \sum_{k \in S} R_k(\theta)$, where

$$R_k(\theta) = y_k \ln(g(T(\mathbf{x}_k))) + (1 - y_k) \ln(1 - g(T(\mathbf{x}_k))).$$

With the softmax activation function $g(\cdot)$, and deviance function, the neural network model is the same as linear logistic regression in the hidden units, and all the parameters are estimated by maximum likelihood. Given the number of hidden units, M , we use gradient descent to minimize $R(\theta)$, called backpropagation, with the update function,

$$\mathbf{b}^{(r+1)} = \mathbf{b}^{(r)} - \gamma_r \sum_k \frac{\partial R_k(\theta)}{\partial \mathbf{b}} \Big|_{\mathbf{b}=\mathbf{b}^{(r)}}, \quad \mathbf{a}^{(r+1)} = \mathbf{a}^{(r)} - \gamma_r \sum_k \frac{\partial R_k(\theta)}{\partial \mathbf{a}} \Big|_{\mathbf{a}=\mathbf{a}^{(r)}},$$

where γ_r is called the learning rate. This gradient is derived from the chain rule of differentiation. (Hastie et al., n.d.) Since the global minimizer of $R(\theta)$ is likely to be an overfit solution, in this project, we use the in-sample -out-sample cross-validation method to train the ANN models and obtain the best M and γ_r .

In this project, `nnet` (Ripley et al., 2022) packages available in R (R Core Team (2022)) are used, which implement a backpropagation algorithm for the prediction of disease prevalence and related statistical values. In brief, backpropagation uses a two-pass algorithm: forward pass and backward pass. In the forward pass, the current weights are fixed, and predicted values are computed. In the backward pass, the errors are computed at the output layer and then back propagated to give errors at the hidden layer units. These error sets are then used to compute gradient descent updates. Also, we apply the `train` function in the `caret` (Kuhn M., 2022) package to determine the number of hidden units, M , and the learning rate γ_r ,

3.4.2 Point and interval estimates

We use the neural network models to estimate the population prevalence. It includes the point and interval estimates. Define $p(\mathbf{x}, \hat{\theta})$ to be the predicted value with the covariate $\mathbf{x}$ and the fitted weights $\hat{\theta}$. After fitting the neural network, we can directly obtain the point estimation for the population prevalence as,

$$\hat{p}_y^{(ann)} = \hat{p}_y^{(p)}(\hat{\theta}) = \frac{1}{N} \sum_{k \in U \setminus S} p(\mathbf{x}_k, \hat{\theta}) + \frac{1}{N} \sum_{k \in S} y_k,$$

where $\hat{p}_y^{(ann)}(\hat{\theta})$ is the population prevalence estimate based on the fitted weights, $\hat{\theta}$.

However, the interval estimation is not straight forward. According to (de Veaux et al. 1998), we first define the residual sum of squares (RSS) overall responses and in the sample as

$$RSS = \sum_{k \in S} \left(y_k - p(x_k, \hat{\theta}) \right)^2.$$

Denote the true weight to be θ_0 , and the true mean of the disease response Y given x to be $\hat{p}_y^{(ann)}(\theta_0)$. The Tylor series expansion of $\hat{p}_y^{(ann)}(\theta)$ at $\hat{\theta}$ is

$$p(x_k, \theta_0) \approx p(x_k, \hat{\theta}) + J(\hat{\theta})(\theta_0 - \hat{\theta}),$$

where $J(\theta)$ is the Jacobian matrix, with i th row to be $\partial p(x_i, \theta) / \partial \theta$, where x_i is the i th observations in S . Thus, the estimated asymptotic variance is

$$\widehat{Var}(\hat{\theta}) \approx \hat{\sigma}^2 \left(J^T(\hat{\theta}) J(\hat{\theta}) \right)^{-1},$$

where $\hat{\sigma}^2 = \frac{RSS}{n-(q+2)M-1}$. Therefore, the $(1 - \alpha)100\%$ Wald confidence interval for $\hat{p}_y^{(ann)}$ is,

$$\hat{p}_y^{(ann)} \pm z_{1-\alpha/2} \widehat{SE} \left(\hat{p}_y^{(ann)} \right),$$

where $\widehat{SE} \left(\hat{p}_y^{(ann)} \right)$ is the estimate of the standard error,

$$\widehat{SE} \left(\hat{p}_y^{(ann)} \right) = \sqrt{\left. \frac{\partial \hat{p}_y^{(ann)}(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}} \widehat{Var}(\hat{\theta}) \left(\left. \frac{\partial \hat{p}_y^{(ann)}(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}} \right)^T}.$$

We compare such a confidence interval with a bootstrapped one, and they are similar.

3.5 Model-assisted ANN with difference estimator

While the model-assisted version of logistic regression is widely adopted, the model-assisted version of ANN is uncommon. We adopt the idea from (Breidt & Opsomer, 2017) to develop a model-assisted version of ANN similar to Montanari and Ranalli(2005).

We first fit the ANN using sample design weights, w_k , which is obtaining $\hat{\theta}^{(d)}$ by minimizing the loss function,

$$R^{(d)}(\theta) = \sum_{k \in S} w_k R_k(\theta),$$

and the $p(x_k, \hat{\theta}^{(d)})$ is the predicted y_k based on the input x_k . According to (Breidt & Opsomer, 2017), we can rewrite $\hat{p}_y^{(p)}(\hat{\theta})$ in Section 3.4.2 as the difference estimator of the population prevalence p_y ,

$$\begin{aligned}\hat{p}_y^{(Diff)} &= DIFF(y; p; \hat{\theta}^{(d)}) = \frac{1}{N} \sum_{k \in U} p(x_k, \hat{\theta}^{(d)}) + \frac{1}{N} \sum_{k \in S} \frac{y_k - p(x_k, \hat{\theta}^{(d)})}{\pi_k} \\ &= \frac{1}{N} \sum_{k \in U} p(x_k, \hat{\theta}^{(d)}) + \frac{1}{N} HT(y - p(x, \hat{\theta}^{(d)})),\end{aligned}$$

where $HT(v)$ is the Horvitz–Thompson estimator of the population total of variable v . This estimator is exactly unbiased, regardless of the quality of the method, since

$$\begin{aligned}E[DIFF(y; p; \hat{\theta}^{(d)})] &= \frac{1}{N} \sum_{k \in U} p(x_k, \hat{\theta}^{(d)}) + \frac{1}{N} E[HT(y - p(x, \hat{\theta}^{(d)}))] \\ &= p_m + p_y - p_m = p_y\end{aligned}$$

Although $\sum_{k \in U} p(x_k, \hat{\theta}^{(d)})$ is not independent of the sample, by similar rationale in Section 3.3, the model variance should be small, and $Var(DIFF(y; m))$ mainly consist of the design variance from the HT estimator:

$$\begin{aligned}\widehat{Var}(DIFF(y; p; \hat{\theta}^{(d)})) &= \frac{1}{N^2} Var(HT(y - p(x, \hat{\theta}^{(d)}))) \\ &= \frac{1}{N^2} \sum_{k, l \in S} \frac{\Delta_{kl}}{\pi_{kl}} \frac{y_k - p(x_k, \hat{\theta}^{(d)})}{\pi_k} \frac{y_l - p(x_l, \hat{\theta}^{(d)})}{\pi_l}.\end{aligned}$$

Therefore, the $(1 - \alpha)100\%$ Wald confidence interval for $\hat{p}_y^{(DIFF)}$ is,

$$\hat{p}_y^{(DIFF)} \pm z_{1-\alpha/2} \widehat{SE}(\hat{p}_y^{(DIFF)}),$$

where $\widehat{SE}(\hat{p}_y^{(DIFF)}) = \sqrt{\widehat{Var}(DIFF(y; p; \hat{\theta}^{(d)}))}$.

4. Simulation settings

To evaluate the performance of the abovementioned methods, we use the R software (R Core Team, (2022)) to perform the simulation. In the study, we aim to mimic the real-world situation, so we generate population data (with a size of 14 million) compatible with the target population from the Canadian Longitudinal Study on Aging (CLSA 2020). We focus on this survey as their sampling design, and population description are well documented. We also generate a stratified sample similar to CLSA's to ensure the assumptions are valid for the variance approximation. Here, we define the strata to be the provinces for simplicity reasons. We computed the survey weights to be proportional to the reciprocals of the individual inclusion probabilities. Details of the population generation and the samples are given in the Appendix.

In the population data generation, we also create gene indicators related to the provinces (Gene1 and Gene2) reflecting the genetic diversity due to immigrants. We also construct the two disease indicators (Disease 1 and Disease 2) related to covariates, sex, smoking status, and gene indicators. We set the proportion of the population having Diseases 1 and 2 to be 25.6% and 36.8%, respectively, to evaluate the effect of the prevalence levels.

As genetic variables are expensive to collect, they are not always available in the health survey. In the simulation, we consider the scenarios in which the gene indicators are available or not.

We vary the sample sizes to observe their effect on prevalence estimation prevalence. For simplicity, we define small, medium, and large sample sizes as 1000, 5000, and 21000 participants in the survey, respectively.

We summarize the details of the simulation settings in Table 1:

Table 1 Summary of the simulation settings

	Disease 1	Disease 2
Population Prevelence p_y	25.6%	36.8%
Genetic variable involved x_3	Gen1	Gen2
Common variables involved x_1, x_2	sex and smoking status	
Sample sizes considered n	1000 (Small), 5000(Medium), 21000(Large)	

For each simulation setting, we iterate 1000 samples and apply the methods mentioned in Section 3 to the generated sample to estimate the population prevalence of Disease 1 and Disease 2. A simple comparison of those methods is presented in Table 2.

Table 2 Summary of the information that various methods can incorporate

Method \ Information	Covariates	Probability sampling design
Direct method	No	Yes
Logistic regression	Yes	No
Logistic GREG	Yes	Yes
ANN	Yes	No
ANN DIFF	Yes	Yes

5. Simulation results and comparisons

We apply the five methods mentioned above for each simulated sample to estimate the population prevalence p_y . Under the i th sample, we obtain the point estimate, $\hat{p}_i$ and the confidence interval $(\hat{p}_{iL}, \hat{p}_{iU})$. We compare the point estimates using biases, and root mean squared error (RMSE), which are defined as

$$Bias = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - p_y), \quad \text{and} \quad RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{p}_i - p_y)^2},$$

respectively. We also compare the confidence interval with the coverage rate and the average width, which are defined as

$$Coverage = \frac{\#\{\hat{p}_{iL} \leq p_y \leq \hat{p}_{iU}\}}{n}, \quad \text{and} \quad Average\ width = \frac{1}{n} \sum_{i=1}^n (\hat{p}_{iU} - \hat{p}_{iL}),$$

respectively. We present the simulation results in Tables 3-6. We boldface the best results for different statistics among the estimation methods.

Table 3 shows the prevalence estimation results of Disease 1, where Gene1 is observable. Among three different sample sizes, logistic regression shows lower bias and high confidence coverage. RMSE values are better in logistic regression and ANN. The Logistic, Logistic GREG and ANN methods have good coverage, but those of logistic and Logistic GREG are closer to the nominal level. If we focus on the average width, ANN outperforms all other methods. The direct method consistently underperforms since it does not consider the covariate information.

Table 3 Simulation results for Disease 1 (prevalence =25.5%) when Gene 1 is observable

Stats/Size	Direct method	Logistic	Logistic GREG	ANN	ANN DIFF
Bias					
Small	-1.1117E-02	6.9449E-04	1.0124E-03	4.7852E-03	-4.1194E-04
Medium	-1.1444E-02	2.0753E-05	2.9901E-04	1.4463E-03	1.4391E-02
Large	-1.1296E-02	-4.3618E-04	-5.5141E-04	2.8883E-04	1.2249E-02
RMSE					
Small	5.5340E-04	2.5865E-04	3.0434E-04	2.3968E-04	4.9452E-04
Medium	2.1451E-04	4.9678E-05	5.6690E-05	5.0287E-05	3.8304E-04
Large	1.4720E-04	1.2259E-05	1.4312E-05	1.2272E-05	3.1943E-04
CI coverage					
Small	8.9900E-01	9.4300E-01	9.5200E-01	9.3900E-01	7.9400E-01
Medium	7.5500E-01	9.4800E-01	9.4700E-01	9.3800E-01	5.5400E-01
Large	2.9400E-01	9.5100E-01	9.5100E-01	9.3900E-01	3.3600E-01
CI average width					
Small	8.0044E-02	6.1911E-02	6.7589E-02	5.5874E-02	5.6835E-02
Medium	3.5797E-02	2.7725E-02	2.9813E-02	2.6408E-02	3.2276E-02
Large	1.7370E-02	1.3446E-02	1.4427E-02	1.3053E-02	1.6328E-02

Table 4 shows the prevalence estimation results of Disease 1 when we do not have Gene1 available. The performances are generally worse than those in Table 3 due to the unobserved variable. Even though we do not have complete covariate information, Logistic, Logistic GREG, ANN, and ANN DIFF outperform the Direct method in all aspects. Logisticals has a more negligible bias for medium and large sample sizes. On the other hand, ANN DIFF has the slightest bias for the small sample as the sample design might provide extra information to the network. RMSE- wisely speaking, ANN performs well for smaller sample sizes, while logistic one is the best for large samples. Logistic regression methods give the best confidence interval coverage. It may be because the simulation follows the logistic regression model. ANN method has the narrowest CI widths, and its CI coverage is satisfactory. ANN could be an excellent option in interval estimation.

Table 4 Simulation results for Disease 1 (prevalence =25.5%) when Gene 1 is not observable

Stats	Direct method	Logistic	Logistic GREG	ANN	ANN DIFF
Bias					
Small	-1.111E-02	7.5651E-04	9.8906E-04	3.5398E-03	-1.8284E-04
Medium	-1.144E-02	4.1462E-05	3.3816E-04	2.2179E-04	7.5634E-03
Large	-1.129E-02	-4.3237E-04	-5.5760E-04	-9.8202E-04	7.0456E-03
RMSE					
Small	5.5340E-04	2.7290E-04	3.1484E-04	2.3881E-04	5.0142E-04
Medium	2.1451E-04	5.1715E-05	5.8937E-05	5.0510E-05	2.1699E-04
Large	1.4720E-04	1.3017E-05	1.5120E-05	1.6100E-05	1.4346E-04
CI coverage					
Small	8.9900E-01	9.4400E-01	9.4700E-01	9.4300E-01	8.2200E-01
Medium	7.5500E-01	9.5300E-01	9.5100E-01	9.3500E-01	7.5900E-01
Large	2.9400E-01	9.5100E-01	9.4700E-01	9.0800E-01	5.0400E-01
CI average width					
Small	8.0044E-02	6.3177E-02	6.8816E-02	5.7776E-02	6.0451E-02
Medium	3.5797E-02	2.8286E-02	3.0404E-02	2.6648E-02	3.3365E-02
Large	1.7370E-02	1.3709E-02	1.4703E-02	1.3119E-02	1.6686E-02

Table 5 shows the prevalence estimation results of Disease 2 when we can observe Gene 2. Disease 2 has a higher prevalence than Disease 1, so comparing Tables 3 and 5 can provide insights into the effects of the prevalence levels. Generally speaking, Table 5 shows better biases and RMSEs as it is easier to estimate a larger prevalence. However, Table 5 also shows wider CIs, as the higher prevalence results in higher standard errors. For the comparison between estimation methods, Table 5 shows ANN DIFF and ANN has better performance in CI coverage, CI width and RMSE. Logistic regression gives better results for bias.

Table 5 Simulation results for Disease 2 (prevalence =36.8%) when Gene 2 is observable

Stats	Direct method	Logistic	Logistic GREG	ANN	ANN DIFF
Bias					
Small	1.6911E-02	7.9496E-04	8.5471E-04	7.7306E-04	3.0567E-03
Medium	1.5929E-02	-3.9188E-05	-2.3413E-04	5.1092E-05	5.5706E-04
Large	1.6343E-02	4.3509E-05	-6.2300E-05	7.6896E-05	2.5517E-04
RMSE					
Small	8.7899E-04	3.3126E-04	3.9027E-04	3.2410E-04	5.0131E-04
Medium	3.6400E-04	6.3095E-05	7.1073E-05	6.3118E-05	8.6753E-05
Large	2.9279E-04	1.4427E-05	1.6526E-05	1.4419E-05	2.7453E-05
CI coverage					
Small	8.9000E-01	9.3300E-01	9.4300E-01	9.1800E-01	8.8200E-01
Medium	6.6400E-01	9.5300E-01	9.5500E-01	9.3900E-01	9.6000E-01
Large	1.1000E-01	9.4400E-01	9.5000E-01	9.3900E-01	9.6200E-01
CI average width					
Small	9.2361E-02	6.7941E-02	7.3759E-02	6.2831E-02	6.9600E-02
Medium	4.1275E-02	3.0435E-02	3.2643E-02	2.9268E-02	3.7376E-02
Large	2.0026E-02	1.4767E-02	1.5812E-02	1.4480E-02	1.8762E-02

Table 6 shows the prevalence estimation results of Disease 2 when Gene2 is not observable. Compared to Table 4, Table 6 shows better biases as it is easier to estimate a larger prevalence. The hidden variable imposes some uncertainty in the estimation, so we have a higher RMSE in Table 5. It also shows a wider CIs, as the higher prevalence results in higher standard errors. ANN method has the best overall performance as it has the negligible biases, RMSE and CI width most of the time. On the other hand, the Logistic GREG and ANN DIFF have better CI coverages.

Table 6 Simulation results for Disease2 (prevalence= 36.8%) when Gene 2 is not observable

Stats	Direct method	Logistic	Logistic GREG	ANN	ANN DIFF
Bias					
Small	1.6911E-02	6.2633E-04	6.2365E-04	-1.3555E-03	1.9023E-03
Medium	1.5929E-02	-2.2823E-05	-2.6748E-04	-2.4673E-05	3.0304E-04
Large	1.6343E-02	3.4194E-05	-1.1945E-04	-9.1609E-04	4.2847E-04
RMSE					
Small	8.7899E-04	3.4447E-04	4.0416E-04	3.1792E-04	5.4096E-04
Medium	3.6400E-04	6.5118E-05	7.3336E-05	6.4729E-05	9.8849E-05
Large	2.9279E-04	1.5115E-05	1.7328E-05	2.0432E-05	5.8805E-05
CI coverage					
Small	8.9000E-01	9.3400E-01	9.3700E-01	9.2600E-01	8.8800E-01
Medium	6.6400E-01	9.4500E-01	9.5100E-01	9.4300E-01	9.4300E-01
Large	1.1000E-01	9.4700E-01	9.4300E-01	8.8400E-01	9.5500E-01
CI average width					
Small	9.2361E-02	6.9454E-02	7.5221E-02	6.3843E-02	7.4230E-02
Medium	4.1275E-02	3.1107E-02	3.3344E-02	3.0168E-02	3.8530E-02
Large	2.0026E-02	1.5093E-02	1.6151E-02	1.4597E-02	1.9220E-02

From the simulation study, it is seen that amongst all the methods applied to the given data, the Direct method performs poorly in all the sample sizes for all the mentioned statistics as it does not consider the auxiliary information, resulting in unsatisfactory results. Its CI coverage decreases as the sample size increases. It is because its asymptotic properties are only valid when the sample-to-population size ratio is small.

Logistic regression is generally good at estimation, mainly when all the variables are observed. Its survey version counterpart, Logistic GREG, has slightly more significant biases and RMSE. However, Logistic GREG is more robust and has better CI coverage when the Gene variables are not observed. It justifies using survey weights in logistic regression in survey data analysis.

Among all the methods, the ANN method generally has a smaller bias, RMSE, and CI widths, but its coverage is generally smaller than the nominal. Its model-assisted version counterpart, ANN DIFF, usually has a larger bias RMSE and CI width as it is more conservative given the sample design. It is because the ANN is trained to have a better prediction power, but the trade-off is having underestimated standard errors and is relatively less robust too.

6. Concluding Remarks

In this study, we have investigated the worthiness of considering the survey design in machine learning for estimating prevalence in a survey study. We have derived the point estimate and confidence intervals of the Direct method, logistic regression method, logistic GREG method, ANN model, and ANN DIFF method.

The simulation study suggests that incorporating the survey design into the neural network algorithm may not be necessary, but it may have better coverage when there is some missing. In terms of point estimation, the neural works outperform the traditional methods. Regarding interval estimation, the logistic regression and logistic GREG have better coverages than ANN. It may be because ANN has an underestimated standard error.

For further studies, we can apply the DIFF method on random forest and k nearest neighborhood method and examine if it could estimate population prevalence accurately and whether survey weights are needed.

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8. Appendix

8.1 The variables in the simulated population:

Smoking

SEX_ASK_TRM

WGHTS_state_TRM

AGE_NMBR_TRM

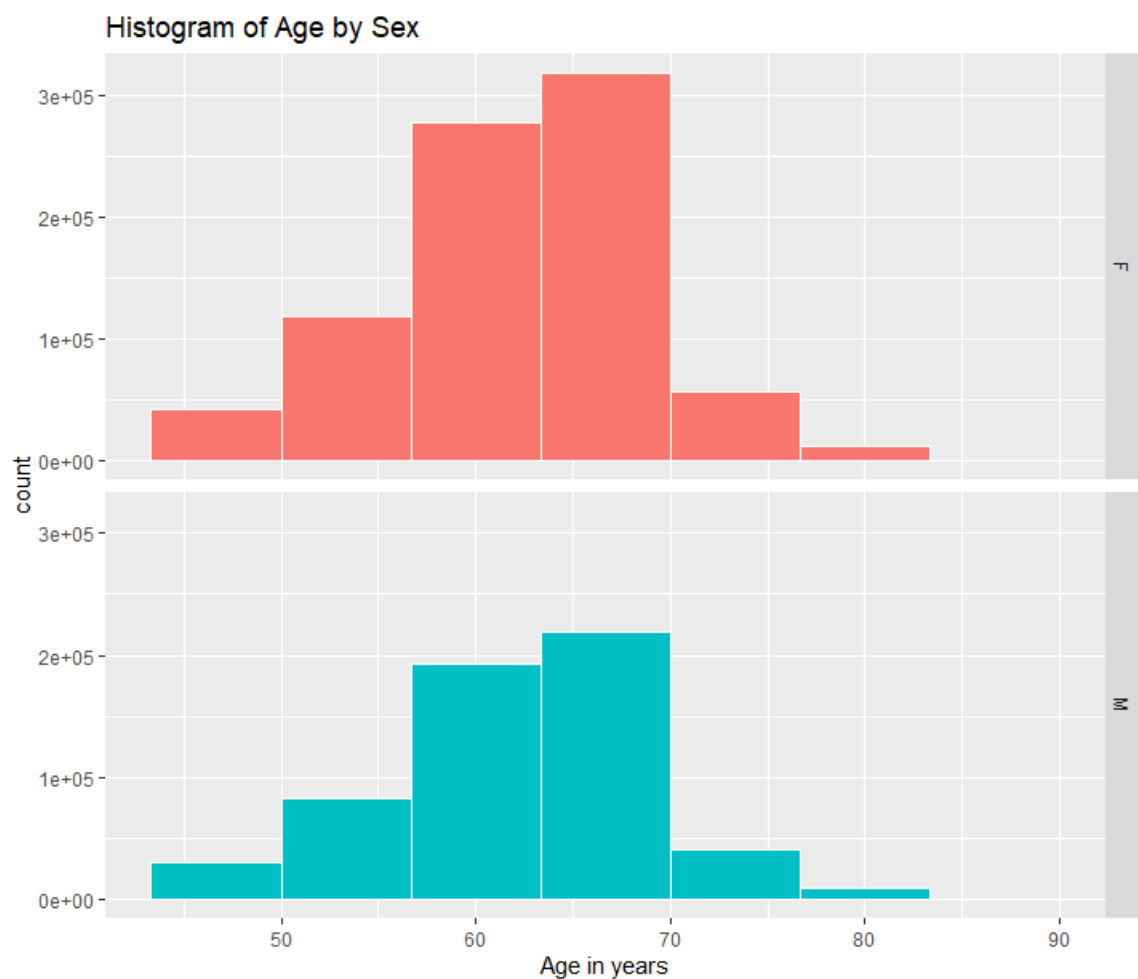
DCS.groups

ED_UDR11_TRM

Disease 1 and Disease 2

Gen1 and Gen2

8.2 Histogram of age by sex



8.3 Tables

Province by Sex

PROVINCE	FEMALE	MALE
AB	112953	78209
BC	125540	87118
MB	51000	35259
NB	22232	15380
NL	17545	12149
NS	25504	17670
ON	130432	90307
PE	4444	3060
QC	318164	220585
SK	19159	13228

Province by smoking

PROVINCE	SMOKING ABSENT	SMOKING PRESENT
AB	154822	36340
BC	179409	33249
MB	73658	12601
NB	32363	5249
NL	23788	5906
NS	35119	8055
ON	192383	28356
PE	6588	916
QC	453970	84779
SK	26663	5724

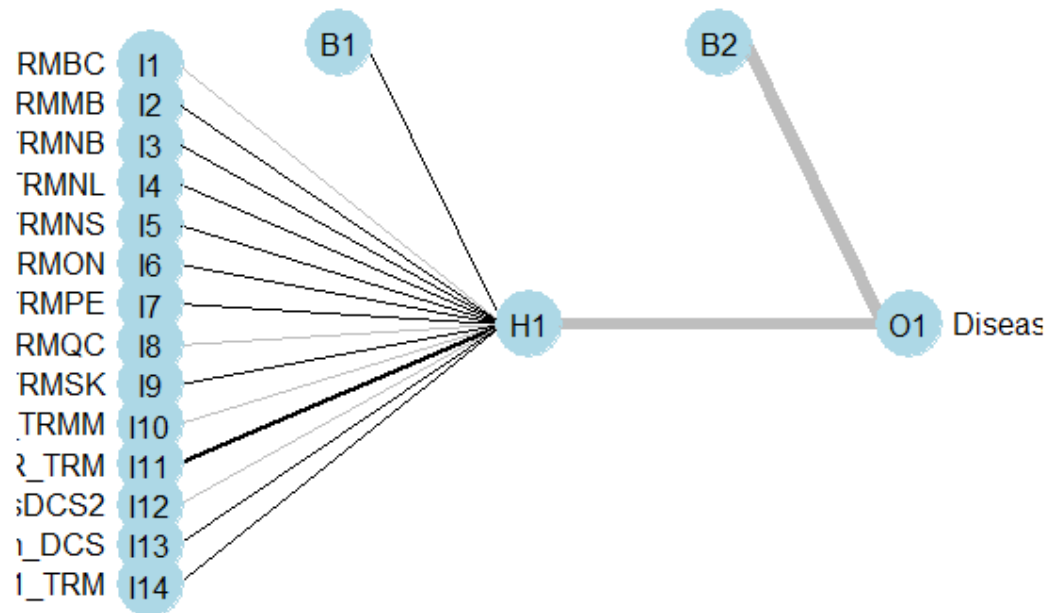
Province by Gen 1

PROVINCE	GENE 1 ABSENT	GENE 1 PRESENT
AB	171718	19444
BC	148818	63840
MB	77605	8654
NB	33888	3724
NL	26774	2920
NS	38911	4263
ON	154346	66393
PE	6793	711
QC	376963	161786
SK	29224	3163

Province by Gen 2

PROVINCE	GENE 2 ABSENT	GENE 2 PRESENT
AB	95969	95193
BC	64110	148548
MB	42941	43318
NB	18836	18776
NL	14908	14786
NS	21639	21535
ON	66436	154303
PE	3725	3779
QC	161776	376973
SK	15998	16389

Nnet plot



8.2 R programs and simulation results

<https://drive.google.com/drive/folders/18ZTRRvJSijNUOZLMjk3tBdwD-ktylOkU?usp=sharing>

<https://drive.google.com/drive/folders/11yoD9ph64fQ09Vjp3bf3mYu5A-XZBuVE?usp=sharing>